

Fullstack Developer Technical Assignment

Overview

Build a scalable real-estate listing platform inspired by:

- 99acres
- NoBroker

The objective of this assignment is to evaluate:

- Fullstack engineering skills
 - Backend architecture
 - Database design
 - API quality
 - Scalability thinking
 - Production-level development practices
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Mandatory Tech Stack

Frontend

- Next.js (App Router preferred)
- Tailwind CSS

Backend

- Node.js
- Express.js

Database

- PostgreSQL

API Documentation

- Swagger / OpenAPI
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Core Requirements

1. Authentication Module

Implement:

- User registration
- User login
- JWT-based authentication
- Protected APIs

Candidate should decide:

- token strategy
 - refresh token implementation
 - middleware structure
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2. Property Listing Module

Authenticated users should be able to:

- Create property listings
- Edit their own listings
- Delete their own listings
- View all listings
- View property details

Candidate should design:

- property schema
 - image handling approach
 - validation strategy
 - ownership handling
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3. Search & Filtering

Implement:

- Search by city/location
- Filter by:
 - budget
 - property type

- bedrooms
- Sorting
- Pagination

Important Requirement

System should support:

- 50,000+ property records efficiently

Candidate is expected to think about:

- database indexing
 - query optimization
 - API response optimization
 - scalable pagination strategy
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4. Similar Properties

Property detail page should display:

- similar properties recommendations

Candidate should decide:

- similarity algorithm
 - optimization strategy
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5. Lead / Inquiry Module

Users should be able to contact property owners.

System should prevent:

- duplicate inquiries
- spam submissions

Candidate should explain:

- validation approach
 - rate-limiting strategy
 - backend protection mechanisms
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6. SEO Optimization

Property detail pages must be SEO-friendly.

Candidate should decide:

- SSR / ISR strategy
 - metadata generation
 - page optimization approach
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7. API Documentation

Swagger/OpenAPI documentation is mandatory.

Expose API docs at:

/api-docs

Frontend Expectations

We evaluate:

- clean UI architecture
 - reusable components
 - responsive design
 - proper loading/error states
 - API integration quality
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Backend Expectations

We evaluate:

- folder structure
 - middleware architecture
 - validation
 - error handling
 - scalable API design
 - security best practices
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Database Expectations

We evaluate:

- schema design
 - relationships
 - indexing strategy
 - optimized queries
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Constraints

Do NOT use:

- Firebase
- Supabase
- no-code generators
- AI-generated boilerplate without understanding